

3.2.4 Scope of Local and Global Variables

دائرہ کار (Scope) بتاتا ہے کہ متغیر کہاں دکھائی دیتا ہے۔

CS XII

Chapter 3: Programming Fundamentals

Lecture 13 of 18

3.2.4 Scope of Local and Global Variables

LEARN IT

Scope is the region of a program where a variable can be seen and used. Python has two scopes you must know for this chapter.

i) **Local scope.** A **local variable** is created inside a function. It exists only while that function is running. The main program cannot read it.

ii) **Global scope.** A **global variable** is created outside all functions. Any function may **read** it, and the main program can use it too.

WORKED EXAMPLE

Local — the name is invisible outside.

```
def calc():
    secret = 50
    print(secret)

calc()
print(secret)
```

OUTPUT

```
50
Traceback (most recent call last):
  File "local_scope.py", line 5, in <module>
    print(secret)
NameError: name 'secret' is not defined
```

First the function prints 50. Then the main program tries `print(secret)` and fails: `secret` was born inside `calc` and died when `calc` finished.

Global — readable inside and outside.

```
x = 10

def show():
    print("Inside function:", x)

show()
print("Outside function:", x)
```

OUTPUT

```
Inside function: 10
Outside function: 10
```

Assignment inside a function creates a new local with the same name. It does not change the global:

```
x = 10

def show():
    x = 99
    print("Inside:", x)

show()
print("Outside:", x)
```

OUTPUT

```
Inside: 99
Outside: 10
```

Inside, `x = 99` is a local. After the call, the global is still 10.

REMEMBER IT

Local = inside the room; global = in the corridor. Everyone can see the corridor. Things left in a room are gone when you leave. Prefer locals. Too many globals make bugs hard to find because any function might be using them.

COMMON MISTAKE

If you only **read** `x` inside the function, Python uses the global. If you **assign** to `x` anywhere in that function, Python treats `x` as local for the whole function. Mixing the two without care causes `UnboundLocalError`. Keep function inputs as parameters and outputs as `return` values instead of editing globals.

اردو میں سمجھیں

دائرہ کار (Scope) بتاتا ہے کہ متغیر کہاں دکھائی دیتا ہے۔ لوکل متغیر فنکشن کے اندر بنتا ہے اور فنکشن ختم ہوتے ہی غائب ہو جاتا ہے؛ باہر سے پڑھنے پر `NameError` آتا ہے۔ گلوبل متغیر تمام فنکشنز سے باہر بنتا ہے اور اندر سے پڑھا جا سکتا ہے۔ فنکشن کے اندر اسی نام کو نئی قدر دینے سے نیا لوکل بنتا ہے، گلوبل نہیں بدلتا۔

CHECK YOURSELF

1. Where is a local variable defined?
2. In the last example, what does the main program print for `x` after `show()` ?

Answers: 1. Inside a function. 2. 10 (the global is unchanged).

Next lecture: 3.3 & 3.4 File Handling and Operations (Modes `r`, `w`, `a`)